**The Hong Kong Polytechnic University**
**Department of Building Environment and Energy Engineering**

**BSE3124 Electricity and Buildings** **2026/27 Semester 1**

# Assignment – Answer Sheets

Group number: _____

Names of group members: _____

_____

_____

_____

**Task 1 – Loading estimation and transformer room location**

Estimation of the building electrical load using CLP COP215 and justifications for transformer room and main LV switch room locations:

**Task 2 – Layout drawings and MCB board schematics**

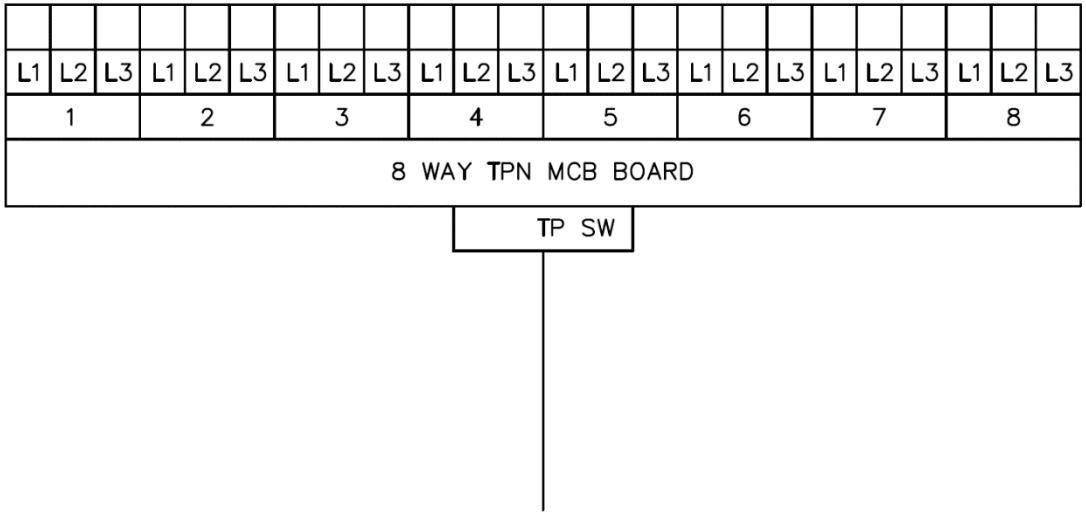
The office tenant chosen for layout design: Office 1 / Office 2 / Office 3

Design considerations of circuit arrangement and protective device selection for small power and lighting layout:

**Task 3 – Maximum current demand calculation and cable sizing**

Maximum current demand calculation:

Cable sizing for the incoming circuit to the distribution board:




MCB BOARD SCHEMATIC

Notes

Client

Project

Drawing

Date

Project number

Drawing number

Scale